

UNIT-3

RELATIONAL MODEL

1Q. Explain About Relational Data Model?

Relational Data Model: The relational model was introduced in 1970 by E. F. Codd. The relational data model is implemented through Relational Database Management System (RDBMS).

- In Relational data model, data is represented in the form of “Tables” or “Relations”. A Table contains rows & columns. Each row is known as “Tuple”. Each column is known as “attribute”.
- We can establish relationship between two tables , but “ a common field “ must exist between two tables.
- SQL Language is used to perform adhoc queries.

Components of Relational Data Model:

- Set of Relations or Tables, tuples, Attributes and Domain.
- Keys in a Relation
- Integrity Rules
- Operations performed on relations or tables.

Relation or Table: A relation is a collection of records. It is also known as “Entity”.

Ex: The records of STUDENT Table are given below

SNo	SName	Addr	DOB
1	DIVYA	KKD	11-nov-2000
2	SAILU	RJY	10-aug-2000
3	SWAPNA	KKD	8-jan-2000

Record: A record is defined as collection of related fields or attributes. A record is also known as “Tuple”.

Examples of record are (1,divya,kkd,.....),(2,sailu,rjy,....)

Field: A Field is defined as a character or a group of characters that has a specific meaning. A field is also known as “Attribute”. Examples of fields are **sno, sname, addr, fname, dob** etc.,

Domain: Domain is a value given to attribute. Example: 1. Male , Female are domains of attribute gender. Example2: 20,24,27 are domains of attribute age.

2Q. Explain About CODD Rules?

Codd's Rules:

Dr. Edgar Frank Codd was a computer scientist. While working for IBM, he invented the relational model for database management in 1970's.

Every database which has tables and constraints need not be a relational database system. There are certain rules for database to be a perfect Relational Database Management System (RDBMS)

- Codd's proposed 13 rules (0 to 12) in 1985 to define a perfect RDBMS.
- If a DBMS needs these rules, it can be called as RDBMS.
- These rules are called as Codd's rules.

Rule 0:

- This is the Foundation rule
- This rule states that any database system must qualify as a relational, as a database and as a management system.
- The other 12 rules are derived from this rule.

The rules are as follows;

Rule 1: The information rule

- A database contains a lot of data i.e; a user data or meta data.
- Every thing in a database must be stored in a table format.
- Each cell should have a single data.
- This rule is satisfied by all the databases.

Rule 2: Guaranteed access rule

- This rule refers to the primary key.
- Each unique piece of data should be accessible by:
Table name + primary key(row) + attribute(column)
- When the combination of these 3 are used, it should give the correct result.

Rule 3: Systematic Treatment of Null

- This rule states about handling the Nulls in the database.
- Null has several meanings i.e; Data is missing cannot applicable or unknown.
- So, null representations must be manipulated by DBMS in a systematic way.

Rule 4: Active online catalog

- This rule illustrate data dictionary.
- Meta data should be maintained for all the data in the database.

Rule 5: Comprehensive data sub language rule

- Any RDBMS database should not be directly accessed.
- It should always be accessed by using a language that supports data manipulation, data definition and transaction management operation.

Rule 6: View updating rule

- Views are the virtual tables created by using queries.
- This rules states that, views are also be able to get updated as we do with its table.

Rule 7: High level insert, update, delete rule

- The database must support set-level inserts, updates, and delete operations.

Rule 8: Physical Independence

- Changes to the physical level (How the data is stored) must not required a change in application.

Rule 9: Logical Independence

- Changes to the logical level (tables) must not be required a change in application based on the structure.

Rule 10: Integrity Independence

- All the integrity constraints can be independently modified without the need of any change in the application.

Rule 11: Distribution Independence

- This rule lays the foundation of the distributed database systems.
- Even if the database is located in different servers the accessibility time should be comparatively less.

Rule 12: Non-Subversive Rule

- If the system provides a low-level interface, then that interface cannot be used to subvert the system and bypass a relational security or integrity constraints.

3Q. What is Relational Algebra? Explain about Relational Algebra operations (OR)

Relational Set Operators?

A. Relational Algebra:

- Relational algebra is a procedural Query language.
- It contains of a set of operations that take one or more relations(tables) as input and produce a new relation as output.

Relational Algebra operations:

The fundamental operations of relational algebra are:

1. select(σ)
2. project(π)
3. union($\cup$)
4. intersect($\cap$)
5. set difference($-$)
6. cartesian product($\times$)
7. rename(ρ)
8. joins($\bowtie$)

1.Select:

- Selection is used to display set of tuples(rows) based on the condition from the relation(table).
- It is a unary operation defined on a single table.
- It is denoted as σ
- It select tuples that satisfy a given predicate or condition

Notation: $\sigma < \text{select condition} > (R)$

2.Project:

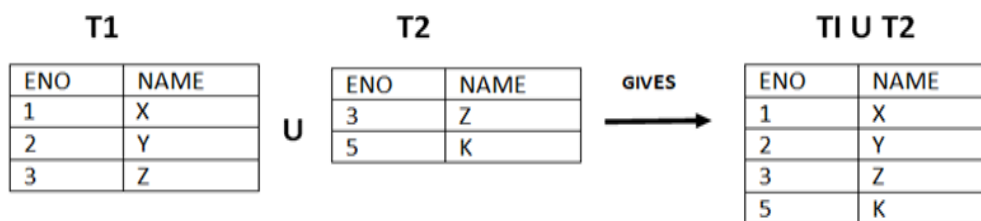
- It is used to display required column data from a table.
- It is denoted as π
- It is a unary operation defined on a single table.

Notation: $\Pi < \text{attributes} > (R)$

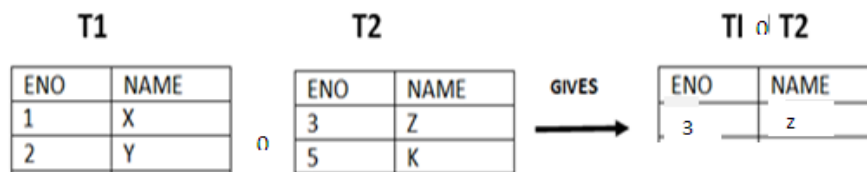
Ex1: $\Pi \text{ eno, name } (EMP)$

3.Union:

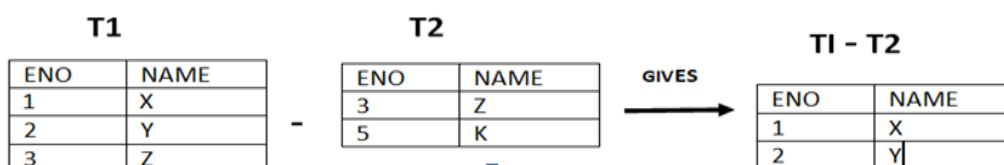
- It is denoted by $\cup$.
- If T1 and T2 are two tables, then it displays all records from both the tables.
- It eliminates the duplicate data.
- It is a binary operation defined on pairs of tables.

**4.Intersect:**

- It is denoted by $\cap$
- It displays only common records from two or more tables.
- It is a binary operation

**5.Set Difference:**

- It is denoted by $-$.
- It displays all the rows in first table which are not in second table.
- It is a binary operation



6.Cartesian Product:

- It allows all the combinations of rows from T1 and T2 in the result set.
- It is denoted by X
- It is also called as cross product.

7.Rename:

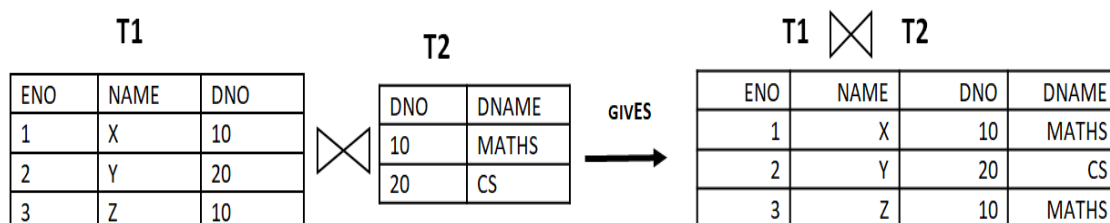
- It is used for renaming tables and attributes.
- It is a unary operation.
- It is denoted by ρ

Syntax: $\rho_S(A_1, A_2, A_3, \dots, A_n)(R)$

- Here, S is new tablename.
- $A_1, A_2, A_3, \dots, A_n$ are attribute names.

8.Join:

- It is used to display data from two or more tables. For Join, both tables must have one common attribute.



- Different types of Joins are equi Join, Self Join, Left Outer Join, Right Outer Join, Full Outer Join.
- It is denoted as $\bowtie$

4Q. Write about Advantages and Limitations of Relational algebra?**Relational Algebra:-**

- Relational algebra is a procedural Query language.
- It contains a set of operations that take one or more relations (tables) as input and produce a new relation as output.

Advantages:-

- It provides a formal foundation for relational model operations.
- The relationship algebra has solid mathematical background.

- It is used as a basis of implementing and optimizing queries in RDBMS.
- It is a high-level language which is in terms of properties of set of tuples.
- Some of its concepts are incorporated into the SQL for RDBMS.
- The usage of relationship algebra of the relation between tables ensures that there is no ambiguity.
- It establishes the links in a complicated network type database.
- The information has to be linked and extracted can be easily manipulated by operators such as project and join.

Limitations:-

- The relational algebra cannot perform Arithmetic operations.
- It can not sort (or) print results in various formats.
- It cannot perform aggregate operations.
- The biggest limitations of relational algebra is that its abstract i.e; we cannot run on it a database server to get a result.
- It only deals with relations.
- The logical and physical modeling will never be completely separable.
- There are simple and natural operations in relations which cannot be expressed by relationship algebra.

5Q. Explain About Relational Calculus?

A: Relational Calculus: Relational Calculus is a non-procedural query language.

- 1 Relational Calculus use “mathematical predicate” to retrieve data from a relation.
- 2 A Predicate is a “Formula or expression used”.
- 3 A relational calculus says “What to do” but never says “How to do”.

Types of Relational calculus:

- 1 Tuple Relational Calculus
- 2 Domain Relational Calculus.

1.Tuple Relational Calculus:

- It was introduced by E.F.Codd in 1972.
- It is a non-procedural query language.

- Tuple calculus is used to select “tuples in a relation based on a condition”.
- Conditions used in tuple calculus are called “Predicated or Well Formed Formulas(wff)”.
- The output is also a “relation”.

It is denoted as

$$\{ t \mid \text{COND}(t) \}$$

Where “t” is a tuple variable.

COND is Condition applied on Relation.

Example1: $\{ t \mid \text{EMP}(t) \text{ and } t.\text{salary} > 10000 \}$

- It implies that the resulting employee tuple will have salary greater than 10,000.
- t.salary is a tuple variable.

Example 2: $\{ t \mid \text{EMP}(t) \text{ and } t.\text{deptno} = 10 \}$

- It displays all the tuples from employee table whose belongs to deptno 10
- t.deptno is a tuple variable.

2. Domain Relational Calculus: Domain calculus was introduced by **Lacroix** and **Pirotte** .

1. It is a non-procedural query language.
2. Domain Calculus used “set of attributes from a Relation”.
3. It is same as TRC but differs by selecting attributes rather than selecting the whole tuples.
4. It is denoted as

$$\{ \langle A1, A2, A3, \dots \rangle \mid \text{COND}(\langle A1, A2, A3, \dots \rangle) \}$$

Where “ $\langle A1, A2, A3, \dots \rangle$ ” are “set of attributes in a relation”.

COND is Condition applied on Relation.

Example1: $\{ \langle \text{eno}, \text{name} \rangle \mid \text{EMP}(\langle \text{eno}, \text{name} \rangle) \text{ and } \text{salary} > 10000 \}$

- It displays eno and ename of employees whose salary is greater than 10000

Example 2: $\{ \langle \text{name} \rangle \mid \text{EMP}(\langle \text{name} \rangle) \text{ and } \text{deptno} = 10 \}$

- It displays name of employees whose belongs to department number 10

6Q.What do you mean by Integrity Constraints? Or Relational Integrity

Integrity constraints :

- These are used to maintain accuracy and correct data in a table.
- Integrity rules are rules enforced(given) on table to maintain database integrity.
- Integrity rules prevent invalid data entry into a table.

There are three types of Integrity Rules:

- 1.Entity Integrity
- 2.Referential Integrity
- 3.Domain Integrity

1.Entity Integrity:-It specifies that there should be no duplicate rows in a table (Primary key and unique)

- **Primary key :** [Primary key](#) uniquely identifies each record in a table. It must have unique values and cannot contain nulls.
- **Unique :**UNIQUE Constraint enforces a column or set of columns to have unique values. If a column has a unique constraint, it means that particular column cannot have duplicate values in a table.

2.Referential Integrity:- It allows to make relationship with other table (Foreign key)

- **Foreign key :** Foreign keys are the columns of a table that points to the [primary key](#) of another table. They act as a cross-reference between tables.

3.Domain Integrity:-It enforces valid entries for a given column(check and not null)

- **NOT NULL :**NOT NULL constraint makes sure that a column does not hold NULL value.
- **Check :**This constraint is used for specifying range of values for a particular column of a table.

7Q. Write about the different types of keys in relational model?**Key:**

A database consists of tables, which consists of records, which further consists of field (attributes).

- A key is a set of one or more columns whose values are unique in a give table.
- A key is a relational means of specifying uniqueness.

Types of keys:

The different types of keys are:

1. Super key
2. Candidate key
3. Primary key
4. Foreign key
5. Alternative key /Secondary key
6. Composite key

1 .Super key:

- Set of all attributes in a relation which identifies data uniquely is known as super key.
- A table can have any number of super keys.
- It is a set of one or more columns to uniquely identity rows in a table.

Ex: Consider student table with attributes sno, sname, marks, address then {sno, sname} and {sname, address} are considered as a set of super keys.

2. Candidate Key:

- It is similar to super key, but does not contains a subset of attributes.
- A super key with no redundant attribute is known as candidate keys.
- So, it is called as a “MINIMAL super key”.

Ex: Consider student table, then

{sno}, {sname, address} are candidate keys.

3. Primary key:

- A primary key is a column or a set of columns in a table that uniquely identifies rows in a table. It is always NOT NULL and UNIQUE.
- Every entity must have a primary key.

Ex: In student table, sno is considered as primary key.

4. Foreign key:

- Foreign keys are the columns of a table that points to the primary key of another table.

Ex: Consider emp and dept

emp(eno, ename, dno)

dept(dno, dname)

Here, dno is the primary key in emp table where as it can act as foreign key in dept table.

5. Alternate key:

- The candidate keys which are not currently selected as primary key are called as Alternate keys /Secondary keys

Ex: Sno is primary key and sname, marks, address are alternate keys.

6 . Composite key:

- If a primary key contains more than one attribute, it is known as composite key.
- **Ex:** {sno, sname} is considered as composite key.

8Q. Write about QBE?

A. QBE:-

- ✓ It was developed in 1970's by Moshe M. Zloof at IBM
- ✓ QBE Stands for "Query By Example."
- ✓ QBE is a feature included in different database applications that provide a user friendly method of running database queries.
- ✓ Typically without QBE , a user must write input commands using correct SQL syntax.If the syntax is incorrect the user maynot get the result.
- ✓ So QBE provides a simple (GUI) "Graphical User Interface" to enter queries and retrieve data from a database.
- ✓ In QBE, instead of writing a query if you specify "Table name,set of attributes,condition" , QBE s/w only will built the query automatically and give result.
- ✓ Using QBE we can apply set operators like union,difference,product,join on a table
- ✓ Using QBE we can also apply database functions like aggregate functions , date and time functions,string functions on table.

- ✓ For example, a user may want to select an entry from a table called "Table1" with an ID of 123.

Using SQL, the user would need to input the command, "**SELECT * FROM Table1 WHERE ID = 123**". The QBE interface may allow the user to just click on Table1, type in "123" in the ID field and click "Search."

- ✓ QBE is offered with most database programs, though the interface is often different between applications.
- ✓ **For example**, Microsoft Access has a QBE interface known as "Query Design View" that is completely graphical.